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RESEARCH ARTICLE

Leveraging Domain-Specific Features for High-Performance Obstacle Classification in Railway Monitoring

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ABSTRACT Railway systems are among the most important transportation routes. Due to their vastness, limited evasion possibilities for trains, and their long braking distances, ensuring the safety of these networks is challenging and can be improved through the application of deep learning. While the localization of objects in the track bed is a common topic, the classification of obstacles is not yet solved robustly for all relevant classes at a high-performance level. This study proposes an architecture tailored to the specific characteristics of track bed images based on a ResNet-50 backbone, leveraging individual characteristics of those images (i.e. high frequency noise, repetitive textures) and significantly reducing misclassifications in this problem domain. Using a five-fold cross-validation and a dataset of 2,003 images in six classes, an average balanced accuracy of 99.26 % can be achieved, outperforming the baseline model (98.50 %) and thereby reducing misclassifications by approximately 46 %. We demonstrate that rather than increasing model complexity, leveraging domain-specific characteristics can lead to highest accuracies. This will enable the application of automatic railway surveillance in the future to ultimately enhance security.

INDEX TERMS Deep learning, object classification, railway, UAV, ResNet, gaussian filter, railroads.

I. INTRODUCTION

Railway networks are among the world's most efficient, safest and most environmentally friendly forms of transportation for large volumes of passengers and goods [1]. The goods transported by train clearly exceed 8 billion ton-kilometers, while the volume of passengers transported is around 3 billion passenger-kilometers worldwide [2]. However, this central role of rail transport in passenger and freight transport can only be reliably fulfilled if safety is guaranteed.

Particularly in comparison to road or air transport, rail transport places special demands on the safety infrastructure: as trains, unlike road vehicles, have hardly any evasion options when confronted with an obstacle and at the same time have very long braking distances of sometimes over a

kilometer, it is crucial to ensure the integrity of the track bed at all times [3], [4]. This integrity consists of two components: the faultlessness of the rails and the appearance of objects in the track bed. This dual problem is further exacerbated by the widespread expansion of rail networks [3]. While the condition of the rails usually only deteriorates gradually over time, objects can appear in the track bed at any time. Continuous monitoring is therefore unavoidable, but is neither possible nor efficient through human monitoring [5].

In the recent past, advances in the field of deep learning have significantly increased the accuracy of automated image processing [6]. Especially when it comes to issues such as the presence of defects or the classification of such defects, these systems now reach human levels and in some cases even surpass them [7]. This makes it possible to analyze large volumes of image data efficiently and automatically, and to assign each image to a specific class.

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To achieve such performance, two options arise: When training from scratch, a model is exclusively trained on the basis of image data from the specific domain [8]. When using transfer learning, a model that has already been pre-trained on large general image data sets is adapted to a new, specific task [9]. Therefore, significantly fewer images from the target domain are required, making transfer learning a particularly promising approach for the surveillance of railway systems.

Such transfer learning approaches have already been used in this domain. E.g., Gao [10] and Zhang et al. [4] show that high accuracies can already be achieved in object recognition, i.e. determining the exact position of objects in the track bed. However, it is questionable whether object recognition is the most suitable approach for the application of such automated inspection procedures: when it comes to the question of whether (emergency) braking or rerouting of the train should take place, it is more important to know the type of object on the track than its exact position. This stems from two arguments: First, automatic detection of the position of the obstacle would only be useful in preparation for automatic removal, which is currently only in the prototype stage [11]. On the other hand, the obstacle type is more relevant, as with certain types of obstacles that can appear large in images (e.g., leaves) trains may be able to continue their journey, while with obstacles such as people, braking is unavoidable. While object detection and segmentation approaches address spatial localization and are thus more complex, our focus lies on classification, which forms the basis of operational decision-making once an object is detected. Only a few studies focus on classification: Chen et al. [12] and Kapoor et al. [13] demonstrate a classification mean precision of 96.88 % and an accuracy of 85.2 % on 3,145 and 749 images belonging to two classes, respectively. By employing a transfer learning approach based on MobileNet, Fayyaz and Johnson [14] achieve an accuracy of up to 88.00 % when utilizing a 3-class dataset of railway crossings.

Building on this existing work, a research gap arises: To the best of our knowledge, there currently is no approach that demonstrates highest classification accuracy robustly for all relevant classes of obstacles. However, such a performance is highly relevant for the practical deployment of such systems. This study addresses the gap by leveraging the characteristic features of this task. The imagery commonly used, primarily captured by uncrewed aerial vehicles (UAV) [15], [16], often suffers from high-frequency noise due to the lower-quality cameras typically mounted on UAVs [17]. Additionally, the aerial views of track beds incorporate repetitive textures, mainly rails and gravel, that can distract deep learning models. To mitigate both issues and achieve high performance robustly for all relevant classes, this work proposes an architecture that integrates a filter specifically designed to lower high-frequency noise and suppress fine-grained textures, namely a Gaussian filter, thereby addressing

the outlined characteristics. Using a 5-fold cross-validation, the approach presented is evaluated on the diverse dataset from Rampriya et al. [16] (six classes, 2,003 images). Our model achieves an average balanced accuracy of 99.26 %, reducing the number of misclassifications by approximately 46 % compared to the baseline model. The main contributions of the work are:

- 1) We propose a tailored deep learning architecture that mitigates the domain specific issues of high-frequency noise and repetitive textures, thereby significantly reducing errors across multiple obstacle types.
- 2) We show that rather than increasing model complexity, a carefully selected preprocessing filter can lead to highest classification accuracies.

The structure of the paper is as follows: The presentation of the related work is followed by the methodology and the results. In the discussion, both theoretical and practical implications, as well as limitations, are pointed out.

II. RELATED WORK

A. CONVENTIONAL RAILWAY OBSTACLE MONITORING

Foreign objects project a huge danger to railway safety. Their intrusion into the rail path can lead to derailments, collisions, delays, and ultimately the loss of life [12], [18]. In relation to the total length of the tracks laid worldwide, however, only a fraction of the rail infrastructure is monitored; if done so, mainly systems preventing signalers or driver errors are employed [3]. Such infrastructure is mainly unmonitored in less developed or very vast countries, with the result that trains have to reduce their speed to such an extent that the route can be monitored by sight. In some cases, passive protection mechanisms such as fences are used in these areas [19]. If real monitoring of the infrastructure takes place, this is often dependent on human components, with people sometimes carrying out inspections by, for example, walking/driving or being required to evaluate previously collected data [20]. Data can be collected via optical camera systems (e.g. on particularly critical structures), vibration detectors [21] or induction loops, whereby a distinction can be made between moving and stationary systems. The former are attached to inspection vehicles or standard trains, for example, while the latter are part of the track structure. The data collected in this way was long evaluated manually; nowadays, mostly mere anomaly detection is used.

Such anomaly detection systems can recognize the presence of an obstacle in principle, but not its type. This makes a differentiated reaction impossible, and in most cases, (emergency) braking or a slow continuation of the journey is initiated, and a decision on how to proceed is made after sight, which can lead to overcautious reactions. This results in a two-fold problem: large sections of routes are unmonitored, while overly cautious reactions occur on monitored sections. Therefore, a practical need for automated, scalable systems to solve this problem exists.

B. EVOLUTION OF MODERN CONVOLUTIONAL NEURAL NETWORKS

While earlier multilayer perceptrons consisted of a few layers of interconnected neurons [22], today's neural networks contain many "deep" layers that enable complex tasks [23]. So-called "convolutional" neural networks (CNN) go one step further and use, for example, trainable filters that lead to even better recognition performance [6]. Such architectures are state of the art today and are used in various domains, from medical tasks [24] to quality control [25] in mass production [26]. The accuracy achieved in image processing reaches and in some cases even surpasses human levels. In order to be able to measure their development in a comparable way, comparative datasets have been developed in the past. These large repositories, which often contain hundreds of thousands of images in hundreds of classes, make it possible to train and test different architectures on a comparable basis [27]. In 2012, Krizhevsky et al. [8] won the first such competition (ImageNet) with their AlexNet architecture, followed by other models such as GoogleNet [28], ResNet-50 [29], or VGG architectures [30] in the years up to the present day. There are two ways to use such deep and comprehensive networks: training from scratch [8] and transfer learning [9]. In the former, a newly initialized network is trained exclusively on the images of the problem, which can lead to overfitting [31], i.e. the network learning the images by memorizing them. In the latter, the architecture is pre-trained on a broad, generalized data set that does not necessarily contain the same problem as the actual application area [32]. The aim is to copy the knowledge acquired through this broad training, apply it and thus achieve high performance without the issue of overfitting. This approach also significantly reduces the dependency on a large amount of training data for the specific problem [33]. Due to this flexibility combined with high performance, transfer learning has established itself as the dominant approach in many domains.

C. DEEP LEARNING IN OBJECT DETECTION AND CLASSIFICATION ON RAILROADS

The use of CNNs to monitor potential obstacles in the track bed has been the focus of research for some time [4], [10], [34]. Three groups have emerged: Object detection, i.e. locating and classifying an obstacle, segmentation, i.e. pixel-accurate detection, and conventional classification. While many papers focus on the first two groups, mere image classification is rarely part of the research in this field.

1) OBJECT DETECTION

Object detection has emerged as the dominant approach for identifying foreign objects on or near railway tracks [4], [15], [35], [36], [37], [38], [39], [40], [41], [42]. Ye et al. [40] achieve a mean average precision of 94.75% with their lightweight model based on 8,776 images in 7 classes (including, e.g., *Pedestrians*) and a fixed train-test split.

He et al. [41] attempt to bring such approaches closer to practice by performing object recognition on images taken in a Chinese subway with obstacles at a realistic braking distance. In this realistic setting, a performance of 93.00 % can be achieved on 6,000 augmented images. Sevi and Aydın [42] and Zhang et al. [4] also address such issues. The former achieves an mAP50 of 88.8 % when detecting *Persons* or *Cars* based on the RailSem19 data set recorded under real conditions. The second uses the largest data set of the studies in this group (12 classes, e.g. *Person*, *Bus*, *Bicycle*, *Motorcycle*, 18,323 images in total) and can achieve a precision of up to 96.6 % (mAP@0.5 = 77.5 %). Although such approaches to object recognition are important for future (autonomous) train systems, they provide information that is not necessary for pure route monitoring in the form of the exact position of the object and, due to this more difficult approach, do not yet reach the level that would be necessary for an area-oriented approach to route monitoring.

2) SEGMENTATION

Segmentation, in which objects are located at pixel level, provides a more precise localization. Some studies lay a basis for the later detection of foreign objects on rails by segmenting the rails from the background. Wang et al. [43] achieve a mean IoU of 89.8 % when segmenting the rails from the background, employing a dataset of 3,000 real-world images. Likewise, Rampriya et al. [44] utilize a drone-based dataset of 2,214 images when additionally segmenting the gauge from the background. The authors achieve a dice coefficient of 0.973 for this three-class task. Other authors build on this basis towards foreign object detection. Chen et al. [12] present an approach in which the authors first segment the rail from the background and then detect potential frame bodies based on YOLO. For this purpose, they employ RAILSegViTNet, a transformer-based architecture. The model evaluated on the well-known RailSem19 can achieve an mIoU of 92.5 % and thus achieve very high performance despite the complexity of the task. Gao [10] choose the most comprehensive approach by segmenting foreign objects in the track bed, such as pedestrians. Based on 5,000 images, an mAP of 98.85 % can be achieved with a fixed split. In general, segmentation approaches are not yet generalized and robust enough to be used for the present task.

3) CLASSIFICATION

Only a limited number of studies approach the problem as a task of image classification or anomaly detection. Wang et al. [45] take a middle way by performing anomaly detection: An autoencoder is trained on 6,656 normal images of the trackbed and tested on 579 images (316 with abnormalities such as phones, bags, etc.), achieving an AUROC of 85.66 %. Fayyaz and Johnson [14] focus on the question of whether certain potentially dangerous objects are present at a level crossing and classify pedestrians,

bicycles, and vehicles. Based on a transfer-learning based model (ImageNet), they re-train using their own dataset of level crossings and achieve an accuracy of up to 88.00 %. Chen et al. [34] address a question very similar to the present study, but with only two classes: The authors classify 3,145 images as intruded (bottles, tires, umbrellas) or non-intruded and are able to achieve a mean precision of 96.88 %. Kapoor et al. [13] address this binary task with 85.2 % accuracy based on 749 thermal images.

4) SUMMARY

In summary, only two papers perform a true classification [13], [34], with both being limited to binary problems. However, a true multiclass classification would be highly relevant: While the ideal case involves identifying both the type and location of an obstacle, classification performance is ultimately more critical than precise localization. Determining the exact position of obstacles becomes relevant only in specific scenarios, such as determining whether the object is within the track bed (which can be addressed by selecting an appropriate image section), if avoidance were possible (as with cars), or for automated removal (a separate issue). Even though object detection and segmentation approaches themselves also include a classification component, their optimization focus is on the spatial localization of objects rather than on distinguishing between potentially visually similar objects. So while object detection and segmentation

approaches are more complex, their classification aspects are often given less consideration, are less finely tuned, and are not optimized for high performance. Our study should therefore be seen as complementary to these approaches.

III. METHODOLOGY

Our study proposes a new architecture to classify all relevant classes of obstacles in railway track beds. To reach this high performance, a Gaussian filter leverages task-specific characteristics. Although this filter is well established in other fields, its systematic integration and evaluation for UAV-based railway imagery represent a targeted and validated adaptation to domain-specific challenges. By applying this model to the UAV-RSOD dataset [16], we significantly reduce misclassifications. We utilize a ResNet-50-based classification model enhanced with data augmentation, filtering, and a custom classification head. To train and validate the filtered and unfiltered model, a two-stage training process with hyperparameter tuning and a 5-fold cross-validation is employed. To mitigate potential overfitting, which is a common problem in such applications, various strategies such as data augmentation, L2-weight decay, early stopping, and dropout layers are used. In this section, the model architecture, followed by the process of training, the methodology of the Gaussian filter, and the evaluation metrics used are outlined.

A. MODEL ARCHITECTURE

1) OVERALL ARCHITECTURE

For this study, a ResNet-based model with data augmentation and a customized classification head with two fully connected layers is used. Two models are trained and evaluated: one unfiltered (*plain*) and one in which the images are filtered using a Gaussian filter. These two models differ only in the application of the filter; architecture and training process are identical. The model structure is shown in Fig. 1. In the first step, the respective input image is resized to the size of 256×256 and a random rotation ($\pm 10^\circ$), a random crop (90%–100 %) and a random translation ($\pm 10\%$) are performed. Such data augmentation strategies can mitigate the risk of overfitting caused by images that are potentially too similar. The Gaussian filter described in section III-B is then used for the filtered model; this step is skipped for the unfiltered model. The image is transferred to the corresponding tensor and normalized with ImageNet values. For both models, unfiltered and filtered, RGB images are utilized.

Afterwards, the image is passed through the so-called stem block, which consists of a two-dimensional convolutional layer, a batch normalization, a ReLU activation function, and a 2D MaxPooling layer.

This block reduces the resolution of the input image and prepares the feature representation for the ResNet-50 backbone. This ResNet-50 backbone is explained in more detail in the following section III-A2. Since the classification head of the ResNet-50 base model is removed, the customized

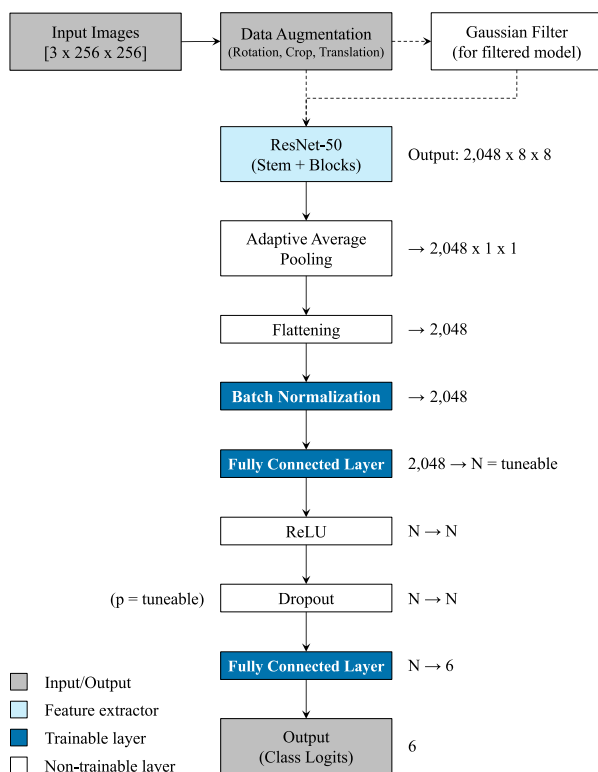


FIGURE 1. Overview of our model. All inputs undergo data augmentation. The dotted paths indicate two variants: direct input to ResNet-50 or preprocessing via Gaussian filtering. Both variants are trained separately.

classification head follows: It starts with an adaptive average pooling layer (2D), which reduces the feature map resulting from the ResNet-50 block ($2,048 \times 8 \times 8$) to the dimension $2,048 \times 1 \times 1$. A flattening layer transforms this input to a one-dimensional vector (2,048) and a batch normalization is applied. The normalized vector then passes through the first fully connected layer, whose number of neurons is tuneable as a hyperparameter. This is followed by a ReLU activation and a dropout layer, whose rate is also tuneable and who can reduce the risk of overfitting. Finally, the second fully connected layer leads to the output by transforming the vector values (dimension based on the neurons of the first fully connected layer) to the output. As the task has six classes, the linear fully connected layer has six neurons. The model uses an Adam optimizer (ASMGrad extension) and is compiled with a cross-entropy loss function.

2) RESIDUAL NETWORKS

The basis of the architecture is a ResNet-50 backbone, pre-trained with ImageNet. ResNet-50 is a deep convolutional network for image classification introduced by He et al. [29] in 2016. The network consists of 50 layers and utilizes a combination of convolutional layers, batch normalizations, ReLU functions, and fully connected layers [46]. The backbone is structured in four stages: (1) three blocks, (2) four blocks, (3) six blocks, and (4) three blocks. Each of these blocks consists of a 1×1 convolution, a 3×3 convolution, and a 1×1 convolution, with batch normalization and ReLU activation being used between each of these convolutions. The main element of the ResNet-50 backbone is the residual connection: For each output from each of the sixteen blocks, the input of the block is added to the output. By doing this, the network primarily learns the differences, which improves the flow of information and stabilizes the training. This significantly reduces the problem of vanishing gradients. The top layer of the ResNet-50 base model is excluded, as it is specific to the dataset used to train the respective model. For initialization, pre-trained weights on the ImageNet dataset were used.

B. GAUSSIAN FILTER

To leverage the characteristics of the given task, we employ a Gaussian filter. Various works have shown that this already brings performance improvements in different domains such as medicine [47] or quality control [48]. In addition, since it is tailored to the specific problem characteristics, the performance of our model is enhanced. A Gaussian filter replaces each pixel with an average sum of itself and the pixels adjacent to it in the x and y directions, weighted with the normal distribution [49]. Here, k pixels are included in the average (kernel size), a $k = 5$ therefore means that the filter extends symmetrically around the respective pixel in the positive and negative x and y directions ($\pm$ two pixels each). The weighting with which these adjacent pixels are included

results from the formula for the normal distribution:

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \quad (1)$$

Here, G describes the normal distribution and σ the standard deviation. In the present study, σ and k were tuned during the hyperparameter tuning. The chosen σ determines the width of the Gaussian curve and therefore the weight with which pixels further away are included into the average. With a larger σ , pixels that are farther away gain weight; with a smaller value, they become less important. As a result, this filter smoothes the image, reducing high-frequency noise, while also conserving important, larger features [49].

C. PROCESS OF TRAINING

In order to train and evaluate our model, a methodologically robust, multi-stage approach was chosen, which is illustrated in Fig. 2. This approach was carried out once for the unfiltered model and once for the filtered model. Five runs are performed for each of the two models in accordance with the stratified 5-fold cross-validation. To ensure the randomness of the approach and data splitting, we set a random seed of 42.

In the first step, the full dataset comprising 2,003 images is divided into five folds (stratified), so each class is equally represented per fold. Then, for each run, a different one of the five folds is defined as a test set, which plays no further role in training. Of the remaining four folds (training set), 10 % are split off as a validation set. A two-stage process then follows, consisting of transfer learning and fine-tuning. In each of the two stages, the actual training is carried out with the remaining 90 % and validated using the split-off 10 %. Hyperparameter tuning is performed for each of the two stages independently, the parameters which are optimized by the tuner and the corresponding values are shown in Table 1. We use the Bayesian search tuner of the Keras library with the aim of minimizing validation loss with a maximum number of 20 trials. For the tuner, we use additional early stopping with a patience of ten, which terminates the trial if the validation loss does not improve for ten epochs. Each trial is conducted with 40 epochs. To ensure that the selected hyperparameter strategy was sufficiently rigorous, runs with 30 and 40 trials were also conducted, using the same methodology as in the main study. Since the mean Balanced Accuracy over all five folds in both runs was lower than that of the main study (e.g., Balanced Accuracy with 30 trials = 98.87 %; Balanced Accuracy with 20 trials (main study) = 99.26 %), the configuration with 20 trials was chosen as the final setting for the main study. This is a computationally efficient choice, without additional effort and without any gain in performance.

In the first stage, transfer learning, the weights of the ResNet-50 backbone are initially frozen and only the custom layers are trainable. Once the hyperparameter tuning is complete, the ideal hyperparameters are used to perform a final training. Here, the ResNet-50 weights remain frozen, and the model is trained with a maximum of 100 epochs and

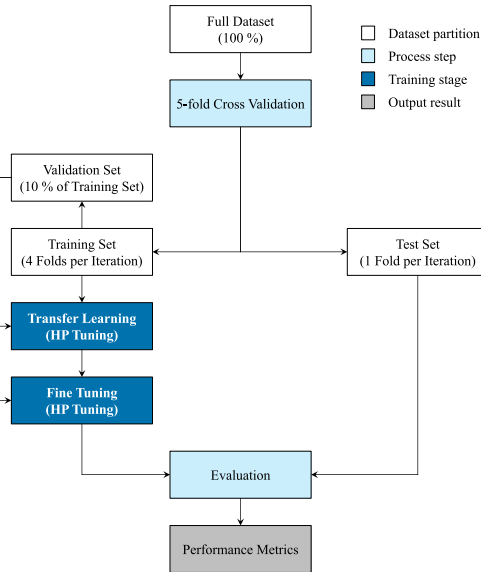


FIGURE 2. 5-fold cross-validation pipeline with transfer learning, fine-tuning, and performance evaluation using validation and test splits.

a patience of ten. If the validation loss has not increased for ten epochs, the training is terminated through callbacks.

TABLE 1. Overview of the hyperparameters used during transfer learning and fine-tuning. The table presents the tuned hyperparameters, including their respective ranges or options. Log-scaled ranges are marked accordingly; step values are indicated where applicable.

Hyperparameter	Minimum Value	Maximum Value	Step / Scale
Transfer Learning Hyperparameters			
Gaussian Kernel Size k	3	11	2
Gaussian Stand. Dev. σ	0.1	2.0	0.1
Learning Rate	10^{-4}	10^{-2}	Log Scale
Weight Decay	10^{-6}	10^{-3}	Log Scale
Dropout Rate	0.10	0.50	0.05
Dense Layer Units	128	1,024	128
Optimizer	AdamW, SGD		
Fine-Tuning Hyperparameters			
Gaussian Kernel Size k	3	11	2
Gaussian Stand. Dev. σ	0.1	2.0	0.1
Learning Rate	10^{-6}	10^{-4}	Log Scale
Weight Decay	10^{-5}	10^{-3}	Log Scale
Dropout Rate	0.10	0.50	0.05
Optimizer	AdamW, SGD		
Batch Size	8, 16		

The model of the epoch with the best results is automatically restored. This approach ensures that only model weights that have led to the lowest validation loss are used, which significantly reduces the risk of overfitting. All regularization strategies were consistently applied across both training stages to ensure methodological comparability and prevent overfitting. The fine-tuning stage then starts, in which the

hyperparameter tuning described above is performed again. This time, however, the weights of the backbone can also be trained, but the selectable learning rates are significantly lower. This hyperparameter tuning is followed by another final train with the same procedure as described above. After finishing both training stages, the evaluation of the models is carried out individually for each fold in order to demonstrate the robustness of the approach. For this evaluation, the fold that was defined as the test set is used.

D. HARDWARE AND COMPUTATIONAL RESOURCES

The experiments were conducted on the HSUPER HPC cluster at Helmut-Schmidt-University/University of the Federal Armed Forces Hamburg, utilizing compute nodes with 256 GB RAM, 894 GB local scratch storage, and two NVIDIA A100 GPUs (40 GB each). Each node is powered by two Intel Xeon Platinum 8360Y processors (72 cores total). A single NVIDIA A100 GPU and a compute node with a 24-hour job limit were allocated, providing sufficient resources. PyTorch 2.5.0, Python version 3.11.7 and Cuda version 12.4.1 were employed.

E. EVALUATION METRICS

To measure performance, the metric definitions provided by `scikit-learn` were used [50]. Eight metrics are used in this study, namely accuracy (Acc.), balanced accuracy (Bal. Acc.), F1-Score, weighted recall/true positive rate (TPR), weighted precision/positive predictive value (PPV), weighted specificity/true negative ratio (TNR), weighted negative predictive value (NPV) and Cohen's Kappa. Accuracy is one of the most widely used metrics and calculates the ratio of correct classifications, i.e. both true positives (TP) and true negatives (TN), to the total sum of predictions, which also contains false positives (FP) and false negatives (FN) in addition to the two mentioned [51]. It is calculated using the following formula:

$$\text{Acc.} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

Although this measure provides an overview of the overall performance of the model, it is sometimes very distorted, especially for imbalanced datasets, as the prevalence of the individual classes is not taken into account. To reduce this problem, the balanced accuracy can be used. This is calculated as the proportion of correct classifications in the total classifications per class i weighted across all classes (N). In multiclass classification, balanced accuracy corresponds to the unweighted average of per-class recalls [50]:

$$\text{Bal. Acc.} = \frac{1}{N} \sum_{i=1}^N \frac{TP_i}{TP_i + FN_i} \quad (3)$$

The proportion of correct positive classifications to true positive samples in a class is measured by the TPR, also known as recall. For multiclass tasks, this value is calculated as a weighted value by weighting by the prevalence. This

is calculated using the following formula, where n_i is the number of samples per class and S is the total number of samples in the task [50], [52].

$$TPR = \sum_{i=1}^N \frac{n_i}{S} \cdot \frac{TP_i}{TP_i + FN_i} \quad (4)$$

Similarly, the TNR can be calculated, which is a metric of the proportion of correctly negative classifications in the true negative samples per class. In the multiclass approach, it is calculated for each class in the sense of a one-vs-rest strategy and also weighted according to its prevalence [50]:

$$TNR = \sum_{i=1}^N \frac{n_i}{S} \cdot \frac{TN_i}{TN_i + FP_i} \quad (5)$$

If the focus is on the correct positive and correct negative classifications in the total positive and negative classifications, the (weighted) PPV and NPV can be used. The PPV calculates the proportion of correct positive classifications in all positive predictions. For multiclass problems, the weighted PPV is often used, which weights this ratio according to the class weights [50], [52]:

$$PPV = \sum_{i=1}^N \frac{n_i}{S} \cdot \frac{TP_i}{TP_i + FP_i} \quad (6)$$

Similarly, the weighted NPV can be calculated, which reflects the prevalence-weighted proportion of correctly negatively classified samples per class [50].

$$NPV = \sum_{i=1}^N \frac{n_i}{S} \cdot \frac{TN_i}{TN_i + FN_i} \quad (7)$$

The F1-score combines these two measures via their harmonic mean. This allows a statement to be made about the quality of the model if false positives and false negatives are equally significant [50], [53].

$$F1\text{-Score} = 2 \cdot \frac{PPV \cdot TPR}{PPV + TPR} \quad (8)$$

Finally, Cohen's Kappa (κ) is a metric for the agreement between the predicted and actual classes that takes into account the possibility of random agreement. A kappa value of 1 would correspond to a perfect match, a value of 0 to a match of exactly the same level as a random match. The calculation is performed using the following formula, where p_e represents the expected random match based on the marginal distributions of the confusion matrix [50], [54]:

$$\kappa = \frac{Acc. - p_e}{1 - p_e} \quad (9)$$

F. DATASET

This study uses a dataset from Rampriya et al. [16]. It contains 2,003 images from six classes (*Iron Rod*, *Boulder*, *Branch*, *Jerrycan*, *Person* and *Barrel*), the distribution of which is shown in the following Table 2.

TABLE 2. Overview of the six classes with descriptions and number of images per class used in this study.

Class	Description	Images
Class 1	Iron Rod	346
Class 2	Boulder	208
Class 3	Branch	548
Class 4	Jerrycan	395
Class 5	Person	234
Class 6	Barrel	272
In total		2,003

This so-called UAV-RSOD dataset was created by the authors by recording videos using a UAV (DJI Phantom 4 PRO UAV, Sony DSC-RXC1RM2, 20MP, full resolution camera), from which the images were then extracted. The track section has a length of 378 m and is located at the junction railway station in Tiruchirapalli, Tamil Nadu, India. During flight, the UAV utilizes a 8.8 mm focal length and 1920 × 1080 HD video recording at 50 Mbps and is flying at 10.5 m above the rail, train, and overhanging lines.

The dataset shows good variance. Table 3 below shows brightness, Michelson contrast, Canny edge density [55], saturation (all on a scale 0-1), and the signal-to-noise ratio (SNR, dB) of all images in the dataset, both as a mean and as a standard deviation (SD). The brightness value was calculated based on the rec.709 standard and shows that the images have a good average brightness of 0.25 and a significant spread of the value with a SD of 0.08. These large differences within the images demonstrate the wide range of the dataset. With 0.90 (mean) and a SD of 0.07, the Michelson contrast also shows the different brightness levels within the images as well as the high variability. The relatively high Canny Edge Density value of 0.32 indicates that the images have a high level of detail, which also shows a certain degree of variability (SD = 0.03). The mean value and SD of the saturation and SNR (0.19 ± 0.04 and 24.7 ± 3.2 , respectively) also demonstrate the variability of the dataset.

TABLE 3. Overview of mean and standard deviation (SD) for brightness, contrast, edge density, saturation, and SNR of all images in the dataset.

Property	Mean Value	SD
Brightness	0.25	0.08
Michelson Contrast	0.90	0.07
Canny Edge Density	0.32	0.03
Saturation	0.19	0.04
SNR [dB]	24.7	3.2

IV. RESULTS

We present the results of our architecture under two configurations: plain model and model with Gaussian filtering. This comparison shows the significant performance gain.

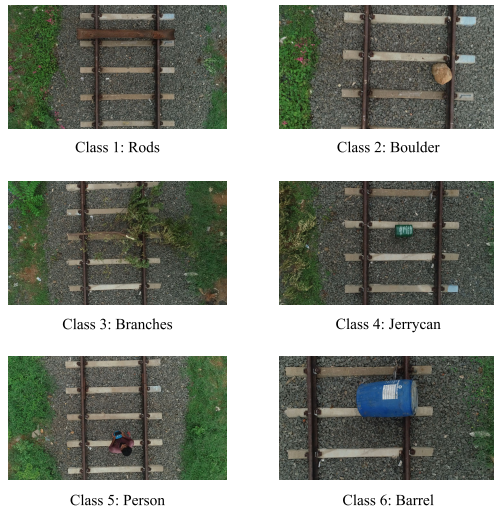


FIGURE 3. Sample images of the six railway obstruction classes: *Iron Rods, Boulder, Branches, Jerrycan, Person, and Barrel*.

Table 4 shows the results of the model without task-specific characteristics being targeted by the filter. The plain model achieves accuracies between 98.00 % and 99.25 % with an average value of 98.70 %. The average balanced accuracy is slightly lower at 98.50 %, with a range between 97.55 % and 99.38 % in Fold V. The TPR and TNR are at an average level of 98.70 % and 99.70 %, respectively, with the highest values of 99.25 % and 99.86 % also occurring in Fold V. The PPV and NPV are on average 98.73 % and 99.74 %, the F1 score is 98.70 % and Cohen's κ is 98.40 %.

TABLE 4. Performance metrics for the model without Gaussian filtering across the five-fold cross-validation (I-V) and their respective averages, Acc., Bal. Acc., TPR, TNR, PPV, NPV, F1-Score, Kappa. The highest value for each metric is highlighted in bold.

Metric	I	II	III	IV	V	Avg.
Acc.	0.9825	0.9900	0.9800	0.9900	0.9925	0.9870
Bal. Acc.	0.9784	0.9900	0.9755	0.9873	0.9938	0.9850
TPR	0.9825	0.9900	0.9800	0.9900	0.9925	0.9870
TNR	0.9953	0.9978	0.9958	0.9976	0.9986	0.9970
PPV	0.9828	0.9902	0.9810	0.9901	0.9926	0.9873
NPV	0.9971	0.9975	0.9962	0.9984	0.9978	0.9974
F1-Score	0.9825	0.9900	0.9800	0.9900	0.9925	0.9870
Kappa	0.9785	0.9877	0.9755	0.9877	0.9908	0.9840

In comparison, Table 5 displays the metrics of the model that leverages the domain features. The accuracy improves to an average of 99.30 % with a maximum of 99.75 % (Fold I). Accordingly, the balanced accuracy also improves to an average of 99.26 %, reaching its maximum in Fold I with a value of 99.69 %. The TPR and the TNR are also significantly enhanced (99.30 % and 99.83 %, respectively), with the latter achieving a maximum of 99.94 % in Fold I. The PPV and the NPV are improved to 99.31 % and 99.85 %, the F1-score to 99.30 %. Finally, the averaged Cohen's κ is increased from 98.40 % to 99.14 %. The low performance variance over all

folds (≤ 0.70 percentage points accuracy range) indicates limited overfitting.

TABLE 5. Performance metrics for the model with Gaussian filtering across the five-fold cross-validation (I-V) and their respective averages, Acc., Bal. Acc., TPR, TNR, PPV, NPV, F1-Score, Kappa. The highest value for each metric is highlighted in bold.

Metric	I	II	III	IV	V	Avg.
Acc.	0.9975	0.9900	0.9925	0.9900	0.9950	0.9930
Bal. Acc.	0.9969	0.9903	0.9914	0.9899	0.9945	0.9926
TPR	0.9975	0.9900	0.9925	0.9900	0.9950	0.9930
TNR	0.9994	0.9979	0.9982	0.9974	0.9988	0.9983
PPV	0.9975	0.9902	0.9926	0.9901	0.9951	0.9931
NPV	0.9996	0.9977	0.9984	0.9978	0.9991	0.9985
F1-Score	0.9975	0.9900	0.9925	0.9900	0.9950	0.9930
Kappa	0.9969	0.9877	0.9908	0.9877	0.9938	0.9914

To demonstrate the effectiveness of our architecture, a Wilcoxon signed-rank test was performed between both models based on ten evaluated folds in two runs each (Fig. 4). Our model demonstrates significantly higher accuracy and a highly significant higher balanced accuracy. The improvement of 0.76 percentage points in the balanced accuracy therefore is significant and consistent over all folds, which is an evidence for genuine enhancement. Furthermore, the corresponding 46 % reduction in misclassifications underscores that even moderate accuracy gains can yield significant practical benefits in high-performance environments.

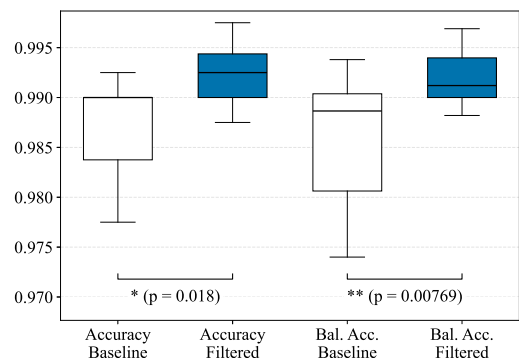


FIGURE 4. Boxplots showing performance of the baseline (white) and filtered (blue) model. Accuracy (left) and balanced accuracy (right) are higher for the filtered model. Wilcoxon signed-rank tests indicate significant improvements for both (* $p = 0.018$ for accuracy; ** $p = 0.00769$ for balanced accuracy (Bal. Acc.)).

Fig. 5 shows the average confusion matrix of the plain model, both in absolute values and in the values normalized to the total number of samples per true class. The baseline model classifies 99.1 % of the *Iron Rods* as such, 0.9 % are incorrectly recognized as *Branches*. 97.6 % of the *Boulders* are correctly classified, 1.4 %, 0.5 % and 0.5 % of these are classified as *Iron Rod*, *Branch* and *Barrel* respectively. Similarly, three error types are shown for *Branches*, of which 98.9 % are correctly classified, 0.5 %, 0.2 % and 0.4 % are incorrectly recognized as *Iron Rod*, *Boulder* and *Jerrycan* respectively. *Jerrycans* show a high performance, only 0.3 %

are incorrectly classified as *Barrel*. 97.9 % of *Persons* in the track bed are recognized as such. Finally, 0.4 % of *Barrels* are classified as *Boulders*, 1.8 % as *Jerrycans*, 97.8 % correctly.

Fig. 6 shows the averaged confusion matrix of the proposed architecture. On average, 99.4 % of the *Iron Rods* are now recognized as such (compared to 99.1 % in the plain model), the performance for *Boulders* increases from 97.6 % to 100.0 %. *Branches* are now recognized as such at 99.5 %, an increase compared to the previously 98.9 %. For *Jerrycans*, the detection rate stays constant at 99.7 %. Images showing a *Person* are recognized by the Gaussian enhanced model at 99.1 % (compared to 97.9 % in the plain model). For *Barrels*, the performance does not increase, remaining at a rate of 97.8 %. This demonstrates that a performance improvement can be achieved for five of the six classes robustly due to the 5-fold cross-validation applied.

Actual	Rod	68.6 (99.1)		0.6 (0.9)			
	Boulder	0.6 (1.4)	40.6 (97.6)	0.2 (0.5)		0.2 (0.5)	
	Branch	0.6 (0.5)	0.2 (0.2)	108.4 (98.9)	0.4 (0.4)		
	Jerrycan				78.8 (99.7)	0.2 (0.3)	
	Person	0.6 (1.3)				45.8 (97.9)	0.4 (0.9)
	Barrel		0.2 (0.4)		1.0 (1.8)		53.2 (97.8)
		Rod	Boulder	Branch	Jerrycan	Person	Barrel
		Predicted					

FIGURE 5. Normalized averaged confusion matrix over all folds for the plain (unfiltered) model. Values represent the absolute number of predictions per class; percentages in parentheses indicate the class-wise accuracy.

To ensure reproducibility, Table 6 shows the final and ideal hyperparameters of the plain model. The learning rate is between $9.9\text{e-}03$ and $5.0\text{e-}04$, the weight decay between $7.1\text{e-}04$ and $1.3\text{e-}06$. The dropout rate for two folds is 0.30, for the others 0.20, 0.40 and 0.50, respectively. The neurons in the first dense layer are between 384 and 768, three times AdamW and two times SGD were used as optimizers. For fine-tuning, the learning rate is stable between $9.9\text{e-}05$ and $4.1\text{e-}05$, the weight decay between $8.8\text{e-}04$ and $3.4\text{e-}05$, the dropout rate varies from 0.20 to 0.50, the batch size used was two times 8 and three times 16. The choice of optimizer is identical to that in transfer learning.

Table 7 shows the values of the filtered model. During Transfer Learning, the Gaussian kernel size varies between 5 and 9, the σ between 0.1 and 1.4. The learning rate and the weight decay are between $6.2\text{e-}03$ and $5.2\text{e-}04$ as well as $9.1\text{e-}04$ and $6.5\text{e-}06$. The dropout rate varies between 0.15 and 0.45, the neurons in the first dense layer between 128 and

Actual	Rod	68.8 (99.4)		0.4 (0.6)			
	Boulder		41.6 (100.0)				
	Branch	0.2 (0.2)		109.0 (99.5)	0.2 (0.2)	0.2 (0.2)	
	Jerrycan				78.8 (99.7)	0.2 (0.3)	
	Person		0.2 (0.4)	0.2 (0.4)		46.4 (99.1)	
	Barrel		0.2 (0.4)		1.0 (1.8)	53.2 (97.8)	
		Rod	Boulder	Branch	Jerrycan	Person	Barrel
		Predicted					

FIGURE 6. Normalized averaged confusion matrix over all folds for the Gaussian filtered model. Values represent the absolute number of predictions per class; percentages in parentheses indicate the class-wise accuracy.

TABLE 6. Plain model: Optimal hyperparameters for transfer learning and fine-tuning across all five folds. Listed are learning rate, weight decay, dropout rate, dense layer size, batch size (for fine-tuning), and optimizer.

Plain	I	II	III	IV	V
Transfer Learning					
Learning Rate	$8.8\text{e-}04$	$9.5\text{e-}03$	$1.9\text{e-}03$	$9.9\text{e-}03$	$5.0\text{e-}04$
Weight Decay	$1.3\text{e-}06$	$1.9\text{e-}05$	$7.1\text{e-}04$	$3.6\text{e-}05$	$1.8\text{e-}04$
Dropout Rate	0.20	0.30	0.40	0.50	0.30
Dense Layer	512	384	512	640	768
Optimizer	AdamW	SGD	AdamW	SGD	AdamW
Fine-Tuning					
Learning Rate	$5.1\text{e-}05$	$9.7\text{e-}05$	$9.9\text{e-}05$	$9.9\text{e-}05$	$4.1\text{e-}05$
Weight Decay	$2.5\text{e-}04$	$3.4\text{e-}05$	$2.5\text{e-}04$	$4.2\text{e-}05$	$8.8\text{e-}04$
Dropout Rate	0.45	0.30	0.20	0.50	0.40
Batch Size	16	16	8	8	16
Optimizer	AdamW	SGD	AdamW	SGD	AdamW

896. The optimizer is chosen as SGD (Fold II) and AdamW (others). For Fine-Tuning, the optimizer is AdamW. In this stage, the kernel size varies between 5 and 11, the σ between 0.1 and 1.6. The learning rate is between $9.9\text{e-}05$ and $2.3\text{e-}05$, the weight decay between $9.7\text{e-}04$ and $3.8\text{e-}05$, the dropout rate between 0.15 and 0.50. The batch size is set at 16.

Fig. 7 shows the training loss and the validation loss from the start of the final fine-tuning until it is stopped by the early stopping algorithm as an example for Fold IV of the filtered model. After an initial sharp decrease, both curves converge and remain at a consistently low level.

To verify the validity of the classifications, we visualized the attention of our proposed model. For this purpose, we used Grad-CAM, which visualizes the gradients of a model and thus provides information about the significance of different

TABLE 7. Filtered model: Optimal hyperparameters for transfer learning and fine-tuning across all five folds. Listed are Gaussian kernel size k , Gaussian standard deviation σ , learning rate, weight decay, dropout rate, dense layer size, batch size (for fine-tuning), and optimizer.

Filtered	I	II	III	IV	V
Transfer Learning					
Kernel Size k	9	5	9	9	7
Stand. Dev. σ	0.1	0.6	1.4	0.7	0.1
Learning Rate	5.2e-04	6.2e-03	9.7e-04	1.1e-03	1.4e-03
Weight Decay	6.5e-06	5.0e-05	9.1e-04	2.6e-05	3.0e-04
Dropout Rate	0.40	0.20	0.45	0.15	0.35
Dense Layer	896	512	256	384	128
Optimizer	AdamW	SGD	AdamW	AdamW	AdamW
Fine-Tuning					
Kernel Size k	11	7	5	5	11
Stand. Dev. σ	0.2	1.6	0.8	0.1	0.1
Learning Rate	9.9e-05	2.3e-05	7.7e-05	8.8e-05	7.1e-05
Weight Decay	9.7e-04	5.3e-05	3.8e-05	5.8e-04	8.8e-04
Dropout Rate	0.50	0.15	0.30	0.45	0.15
Batch Size	16	16	16	16	16
Optimizer	AdamW	AdamW	AdamW	AdamW	AdamW

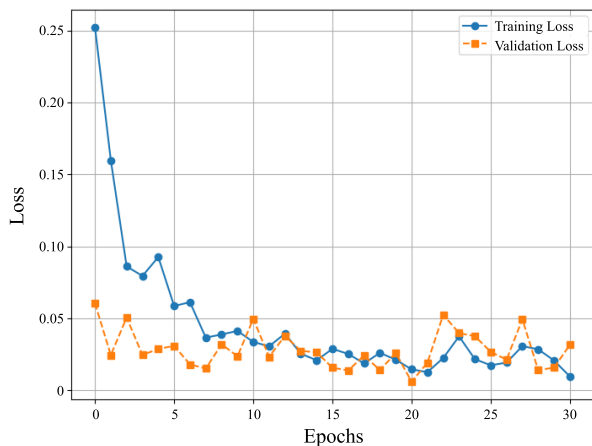


FIGURE 7. Training (blue) and validation loss curves (orange) during fine-tuning for Fold III of the filtered model. The training loss decreases steadily, while the validation loss remains consistently low and stable demonstrating effective early stopping.

image areas for classification. The results are shown in Fig. 8 as examples for one image per class, with red representing more relevant areas and blue representing less relevant areas. All predictions were correct and based on the relevant image features. For the class *Rod*, the rod lying diagonally in the image is correctly identified as the most relevant feature, while for the class *Boulder*, the boulder in the upper center of the image is identified as the most relevant feature. The branch located on the right edge of the image was also correctly considered highly relevant, as was the jerrycan located in the upper area. For the class *Person*, the person in the lower left part of the image is decisive, and for the class

Barrel, the barrel located in the upper part of the image is decisive.

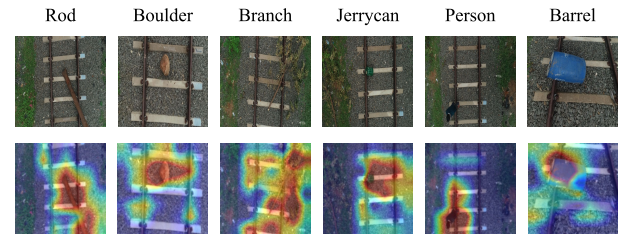


FIGURE 8. Grad-CAM visualizations for all classes (*Rod*, *Boulder*, *Branch*, *Jerrycan*, *Person*, *Barrel*). Top: original images. Bottom: heatmaps highlighting model attention (red = high relevance, blue = low relevance).

V. DISCUSSION

A. CONTEXTUALIZATION WITHIN EXISTING LITERATURE

The literature regarding this problem area to date has focused predominantly on the localization (segmentation [12], [43], [44], object detection [4], [15], [35], [36], [37], [38], [39], [40], [41], [42]) of obstacles in the track bed, and less on the significance of the obstacles in terms of content. If such classification tasks are addressed, this is not done for the necessary breadth of all practically relevant obstacles, but mostly for binary tasks [13], [34]. The present study shifts this view and shows that the key to high classification accuracy for many relevant classes lies in the use of problem-specific filters. The average balanced accuracy achieved in this way thus represents a new, high-performance benchmark in a relevant but currently poorly addressed field of research. The performance value achieved by the architecture used (balanced accuracy of 99.26 %), which is tailored to the characteristics of UAV images of the track bed, is above the previously achieved mean precision of 96.88 % by Chen et al. [34] and the accuracy of 85.2 % by Kapoor et al. [13], although a wide range of possible obstacles was addressed for the first time. Our perspective thus complements that of existing research.

B. TYPES OF ERRORS

Our proposed model solves the problem addressed with an average balanced accuracy of 99.26 % and thus at a very high level and 0.76 percentage points better than the baseline model (98.50 %). Although there is an imbalance of images per class (e.g., *Boulder* as the rarest class with 208 images, *Branches* with 548 images most frequently represented), there is no systematic imbalance in the number of misclassifications depending on the number of images per class. The model thus appears to generalize well within the given unbalanced dataset, which is a promising capability for practical application where access to a large number of sample images for rare obstacle types is unlikely.

Misclassifications usually only occurs with a single image per error type (0.2 misclassified images per error type on average), which is why no systematic errors or distortions can be assumed. Only *Barrels* are misclassified as *Jerrycans*

five times as much, with 1.0 misclassified images per fold on average. One reason for this may lie in the similarity of the objects: Both objects are cylindrical obstacles and thus clearly differ from elongated *Iron Rods* or *Branches* and from on aerial images more circle-shaped *Boulders* or *Persons*.

In addition to this individual analysis, the overall error pattern can also be considered. Noticeably, the detection performance across the classes tends to decrease. While 99.4 % of actual *Iron Rods* are correctly recognized for class 1, only 97.8 % of actual *Barrels* are recognized as such. This observation may be an indication that classes are arranged by complexity: The recognition of *Boulders* or *Iron Rods* is significantly easier for the model than that of *Persons* or *Barrels*. The key observation arises in a second step: the majority of misclassifications lies in the lower triangle of the confusion matrix, indicating that the model tends to categorize objects that are more difficult to detect into simpler classes and thus avoids riskier classifications. However, this type of behavior could be potentially dangerous in safety-critical applications, as the model must not tend to classify objects into “simpler” classes.

C. COMPARISON TO OTHER MODELS

The model we propose is based on a ResNet-50 backbone. However, other backbones have been tested to identify the best performing baseline model before evaluating the performance of the enhanced model. Three other CNNs, namely DenseNet-121, EfficientNet-B4 (Eff.Net-B4) and MobileNetV2, as well as a visual transformer, specifically the Swin-V2 shifting window transformer, were tested for this purpose. While the former were to be tested as conventional CNN backbones as direct benchmarks for ResNet, Swin-V2, as a visual transformer, has shown high performance in rail inspection tasks in recent research. For example, the RailFormer architecture proposed by Guo et al. [56] used a global criss-cross attention mechanism to improve object detection, while Zhao et al. [57] use multi-level attention blocks within the Swin Transformer in their CBAM-SwinT-BL framework. The RailTrack-DaViT investigated by Phaphuangwittayakul et al. [58] employs a dual-attention ViT architecture and achieves accuracies of up to 99.2 %.

The average accuracies of these approaches, evaluated using the same methodological procedure as described in section III (5-fold cross-validation, hyperparameter tuning), are shown in the following table 8.

TABLE 8. Comparison of the average accuracy (Acc.) across different backbone architectures (EfficientNet-B4, DenseNet-121, MobileNetV2, Swin-V2) using 5-fold cross-validation.

Metric	Eff.Net-B4	DenseNet-121	MobileNet-V2	Swin-V2
Acc.	0.9810	0.9850	0.9770	0.9800

Using an EfficientNet-B4 extractor, an average accuracy of 98.10 % was achieved, compared to a value of 98.50 % for DenseNet-121. When using MobileNetV2, this value is 97.70

%, and when using a Swin-V2 transformer, the value reaches 98.00 %. The ResNet-50 backbone we chose in the baseline model is therefore the ideal backbone for the present case, with an average accuracy of 98.70 % (see Table 4).

D. ABLATION STUDY ON IMAGE PREPROCESSING FILTERS

To investigate whether other preprocessing techniques might achieve equal or better results and if the observed performance increase can be attributed to specific characteristics of the Gaussian filter, a second ablation study was conducted. In addition to the Gaussian filter used in the proposed architecture, the median filter, the bilateral filter, and histogram equalization (HE) were also tested. All filters were integrated into the architecture individually using the same methodology as in the main investigation (see chapter III) and were also tested using cross-validation, hyperparameter tuning, and fine tuning to ensure methodological comparability. Table 9 summarizes the Bal. Acc. of the architecture we proposed (Gaussian) and the three additional filters tested.

TABLE 9. Comparison of the average balanced accuracy (Bal. Acc.) across different preprocessing methods.

Filter	Gaussian	Median	Bilateral	HE
Bal. Acc.	0.9926	0.9868	0.9838	0.9890

The architecture we proposed shows an average Bal. Acc. of 99.26 % across all folds, outperforming all other tested preprocessing methods. The second-best result was achieved with HE (98.90 %, global equalization on the Y channel), followed by the median filter (98.68 %, size = 5) and the bilateral filter (98.38 %, $d = 9$, $\sigma_{\text{color}} = 75$, $\sigma_{\text{space}} = 75$). The Gaussian filter achieved the highest balanced accuracy, which shows that the filter is particularly well suited to the problem at hand.

E. GAUSSIAN FILTER

1) DOMAIN-SPECIFIC ADAPTION OF THE GAUSSIAN FILTER
The Gaussian filter used in our model is a preprocessing filter already established in image classification, which shows performance improvements in various domains [48]. However, just because such an architectural component is established does not mean that its use is necessarily trivial; rather, it can be a deliberately chosen, systematically evaluated, domain-specific choice. Our work shows that this filter brings statistically highly significant improvements in performance compared to unfiltered models, especially for railway data, which are not to be expected in other domains: Drone images of railways in particular exhibit special characteristics, such as high-frequency structures [17] in combination with gravel, iron rails, and adjacent vegetation, which are ideally addressed by the smoothing properties of the Gaussian filter. In other domains where high-frequency properties in the image are particularly informative (e.g., in the medical field such as histopathology or in industrial inspection of fine structures), such a preprocessing choice

is not appropriate. The choice of Gaussian filtering for the specific task of railway obstacle detection is therefore not arbitrary, but explicitly domain-specific and represents an architectural decision validated by our methodology.

Additionally, when compared to other domains in which UAV-based inspections are carried out, the Gaussian filter is disproportionately well suited to the problem at hand. When UAVs are used to monitor agricultural land [59] or forest areas [60] for weeds, diseases, or vegetation structures, for example, the problem usually lies in classifying images with larger, more homogeneous areas such as planted fields or woodlands. In contrast, the task of inspecting railways is characterized by clearly distinguishable, significantly more contrasting and noisy areas such as gravel areas in the middle of the image, iron rails, and smaller vegetation. Since relevant obstacles are usually located in the track bed and thus directly on the area with a high-frequency pattern, the Gaussian filter is also disproportionately well suited to the problem at hand compared to other domains.

2) IMPACT OF THE GAUSSIAN FILTER

Part of the architecture presented in this paper is the leverage of problem-specific features such as disturbing recurring textures or high-frequency noise and thereby solving the problem by using a Gaussian filter. An analysis of the different performance of the baseline and the filtered model can provide an empirical assessment of the filtering effect. In general, performance was improved on average and various misclassifications were reduced. The most significant reductions are *Boulder* recognized as *Iron Rod*, *Person* recognized as *Iron Rod* and *Person* recognized as *Barrel*. This indicates that classes that were previously frequently confused despite clear shape differences, such as round structures like *Boulders* and *Persons* compared to elongated shapes like *Iron Rods*, are now recognized more reliably. This could be an indication that the Gaussian filter directs the focus of the model more strongly on such shape features and thus reduces interfering influences.

Conclusions about the effectiveness of the filter can also be drawn from the misclassifications occurring in the filtered model: by far the most frequent error is the misclassification of *Barrels* as *Jerrycans*. *Jerrycans* and *Barrels* differ in the data set used, apart from their color, mainly in fine textures such as inscriptions and surface structures, while they are very similar in their cylindrical shape. The frequent confusion between these two classes could indicate that the Gaussian filter smooths these fine granular texture features as intended and thus reduces their distinguishability.

As well as having a positive impact on detection accuracy, the properties of Gaussian filtering may also have potential advantages for other aspects of monitoring critical infrastructure. If the focus is not on objects accidentally placed in the track bed but on potential sabotage, the possibility of so-called adversarial attacks must be taken into account when using such systems [61], [62]. Although adversarial robustness was not empirically evaluated in this

study, it represents an important consideration for future work. Such attacks are characterized by the deliberate deception of automatic (image) classification systems with the aim of causing misclassifications in order to trigger potentially dangerous consequences [63]. This effect is caused, for example, by the addition of large interference artifacts, but also by the use of specific patterns that can significantly interfere with recognition [64], [65]. In relation to the classification of objects in the track bed, potentially dangerous, larger or heavier objects could be deliberately manipulated in order to recognize them as smaller or less dangerous objects. The use of a Gaussian filter shifts the focus from local, fine-grained textures to larger silhouettes, which, as demonstrated by Naseer et al. [66], may reduce the impact of high-frequency adversarial perturbations. This potential effect, however, remains to be empirically validated in future works.

F. DEPLOYMENT & REAL-TIME PERFORMANCE

Since our model is trained and evaluated on a HPC cluster but ultimately aims to form the foundation for potential deployment in resource-constrained environments, we evaluate its inference performance on a less powerful system. To approximate, though not fully replicate, embedded scenarios, we measured inference performance on a consumer-grade device (macOS 15.6, Apple Silicon, 6 CPU threads, no GPU acceleration) as shown in Table 10. A batch size of 1 and an image size of 256×256 pixels were used with varying numbers of CPU threads, and the median latency (forward-only, end-to-end including preprocessing) and throughput were measured in each case.

TABLE 10. Median inference latency (p50) and throughput for CPU thread sweep without pruning (batch size = 1, image size = 256×256).

Threads	Forward p50 [ms]	End-to-End p50 [ms]	Throughput [img/s]
1	60.28	73.01	12.94
2	67.31	62.43	13.40
4	57.72	56.52	14.78
6	71.92	61.42	12.81

The results show a median end-to-end latency of 73.01 ms and a throughput of 12.94 img/s for a single CPU thread. The optimum is four CPU threads with a median end-to-end latency of 56.52 ms and a throughput of 14.78 img/s. Based on these results, real-time scenarios in the range of approx. 12-15 img/s are realistic. Assuming an image coverage of 4 m and an overlap of 25 %, our architecture can cover approximately 36-45 m/s, which matches typical drone flight scenarios. The findings suggest that, under approximated conditions, our model could potentially meet real-time requirements. However, further validation on dedicated embedded hardware is required to investigate its practical deployability. Nevertheless, techniques such as pruning or quantization remain optional for deployment on severely resource-limited embedded hardware.

G. PRACTICAL IMPLICATIONS

The present study addresses a problem that has so far been less in the focus of the literature through the task of classifying obstacles in the track bed at a high performance level for a broad number of classes. The practical implications arise directly from the high model performance achieved: the use of automated obstacle classification systems in the track area requires the highest possible performance for all relevant obstacle classes. Regardless of the implementation of the system used, which is discussed in the next paragraph, only a limited area of track can be recorded per image. In the case of comprehensive monitoring of a large railway network [3], this leads to millions of individual images and even initially seemingly small improvements in performance of tenths of a percentage point can make a large difference between stable operational reliability and a number of misclassifications with resulting faults.

By proving that the classification of obstacles can be approached at a high performance level in an experimental setup, the second step is to define the location of the integration of such a system. Three possibilities exist. Firstly, such systems can be integrated directly into trains [67]. The main issue is the sensory input: unlike aircraft, trains are characterized by significantly less limited installation space and fewer restrictions in terms of payload, and a processing unit required for the use of the present model would not be significant in relation to either the overall weight or the overall performance of typical train systems. However, since the goal of the approach is to reduce accidents [3], [67], the integration of the sensor system into the train must ensure that obstacles are classified early enough to enable emergency braking. Since braking distances are often several hundred meters to several kilometers long [68], either due to the high total weight of the train or the high total mass, the sensor technology used would have to be extremely high-resolution. Even if such a system were integrated, curve guidance, the influence of light from oncoming trains or insufficient long-range illumination can significantly reduce detection performance, making this integration option impractical. The second option is to install the camera systems directly along the track, which is nowadays mainly implemented at level crossings and in stations [69], [70]. Such a system has no direct restrictions in terms of weight and power consumption of the units, and centralized image processing can take place to enable an economical system design. In addition, only static images are produced, which tend to be captured by high-quality cameras and contain only a few features that interfere with image recognition. However, such systems must be installed with a great deal of structural effort, and the maintenance of this static system is also time-consuming. Finally, there is a risk of sabotage, which is why this option is also not ideal. The third option is the use of separate, moving detection technology such as UAVs [15]. This technology has decisive advantages: compared to the option just discussed, there is no need for cost-intensive equipment of the entire route with camera systems, and

relevant sections can always be monitored on an ad hoc basis. Compared to the use of camera technology directly on the train, drones can implement predictive route monitoring, enabling this data to be used for emergency braking, for example. This makes drones the ideal and widely used image capture technology and the classification models used must deal with the disadvantages of this capture technology. These include two in particular: due to the limited installation space and the restricted payload, only cameras of limited quality can be used, which tends to result in noisy images. In addition, the movement of the drone can lead to motion blur, which further impairs the classification performance [17]. The Gaussian filtering used in this study counteracts precisely these two effects by reducing image noise and emphasizing uniform structures. As a result, the presented architecture shows promising characteristics for this technically challenging application scenario.

VI. CONCLUSION

Railways are among the most popular and extensively used means of transportation worldwide [1], which means that there is a special responsibility for their safety. Current approaches focus mainly on determining the position of obstacles in the track bed and do not address classifying all relevant types of obstacles. This study addresses this gap by leveraging individual characteristics of track bed images (i.e. high frequency noise, repetitive textures) using a ResNet-50-based architecture enhanced with a Gaussian filter. This approach smooths such high frequency noise and shifts the attention of the CNN from such repetitive textures towards the shape of the obstacles. This tailored approach was evaluated on a dataset comprising 2,003 images in six classes using a stratified 5-fold cross-validation. By reaching an average balanced accuracy of 99.26 %, our proposed model significantly reduces misclassifications by approximately 46 %, demonstrating a statistically verified improvement over the baseline. This value also outperforms the performance of the unfiltered model (98.50 %).

Thereby, our study does not only propose a new deep learning architecture that mitigates the domain specific issues of high-frequency noise and repetitive textures, significantly reducing misclassifications when classifying multiple obstacle types in track beds, but also shows that rather than increasing model complexity, a carefully selected preprocessing filter can lead to highest classification accuracies. This twofold contribution makes a valuable contribution to the existing literature in this field. Our findings have theoretical and practical implications. First, we demonstrate that leveraging task specific characteristics by a preprocessing filter can enhance model performance significantly. Second, our study demonstrates the ability of modern CNNs to robustly classify obstacles in the track bed for a broad range of relevant classes, paving the way toward automatic railway monitoring and thereby enhanced operating security.

A. LIMITATIONS

The limitations of this study concern both the proposed model and the dataset used to validate the approach.

1) LIMITATIONS OF THE DATASET

The dataset used for validation by Rampriya et al. [16] is, to the best of our knowledge, currently the dataset with the most diverse classes of obstacles for UAV-based classification of railway obstacles. Nevertheless, it has various limitations, from the location and type of recording to weather conditions and obstacles. First, the dataset does not fully comprise all relevant classes of obstacles; for example, *bicycles* and *vehicles* [4] are missing. This limits the current results regarding their generalizability. In addition, the objects shown are all visible without cover or other interfering influences, and there are no overlaps, which also overestimates the performance of the model. In addition, there are limitations regarding the generalizability of the results for the existing classes: Since obstacles can occur on different sections of the railroad system, including e.g. special sections such as level crossings, the results based on the existing dataset, which only uses a limited stretch of railroad, are not fully generalizable. Furthermore, no diversity in terms of geography and infrastructure is taken into account, as all images originate from a specific section in India. Different surfaces, rail types, or fastenings are not sufficiently considered. Ultimately, the dataset does not include any images taken in other weather conditions or at other times of day, which would be necessary for practical application. Extreme scenarios such as snow, fog, or images that look different due to lights are not included. In summary, the generalizability and robustness of the model are limited: Since the approach focuses on the broadest possible classification of obstacles, the robustness of the model cannot be conclusively assessed with the currently available data. Future data sets must therefore not only cover a wide range of classes, but also include more diversity in terms of infrastructure, weather, and geography. Only with such availability can proposed approaches in this domain be realistically tested and their robustness ensured.

2) LIMITATIONS OF THE PROPOSED MODEL

The second group of limitations relates to the model used, which focuses entirely on optical, static image data and classifies it using a CNN. As a result, it is subject to limitations in both respects. The reliance on optical images does not include all available information about the obstacles but only their (two-dimensional) shape and color features. Potentially important other features such as height, three-dimensional shape or others cannot be included, which limits performance [38]. In addition, sequential processing of multiple temporally linked images or a video could provide a better understanding of the obstacle, which could be enabled by recurrent CNNs. However, even if such an architecture were proposed, there is currently no comprehensive dataset

to evaluate it. Last, there is a risk of overfitting. Even though various measures have been taken to prevent this, such as cross-validation, data augmentation, L2-weight decay, early stopping, and dropout, there is still a residual risk. All conclusions regarding the model's performance and its deployability in practice should therefore be viewed with the necessary caution.

B. FUTURE WORK

Based on the limitations outlined above, opportunities for further research arise. Regarding the dataset, the scope of available data should be enhanced. This applies to the number of object classes, number of images per class, type of track system, locations of obstacles on the tracks, weather conditions, and time of day. Various promising possibilities could arise in the near future, especially for the problem domain at hand. As more and more railway providers utilize UAVs in pilot projects for, e.g., maintenance tasks, there will be a faster-growing basis of data than in recent years. Additionally, as generative adversarial networks have significantly improved in terms of performance, issues regarding different daytime settings [71] or weather conditions [72] can potentially be targeted using synthetic data augmentation. The developed models for classifying UAV images should furthermore be transferred more extensively between various domains of monitoring to improve generalization. Since a broader range of UAV-based datasets exists, future research could test the applicability and robustness of approaches from other domains, such as road safety or infrastructure monitoring, for the domain of railway safety. Ultimately, future works should account for the robustness of such models against adversarial attacks [61], [62]. Since railway safety systems are not solely responsible for ensuring security against randomly occurring obstacles but also against deliberate attacks, such attacks will occur as soon as CNN-based monitoring systems are broadly used.

In summary, our work widens the focus of existing research from object detection and segmentation towards high performance object classification. Thereby, we propose valuable contributions toward the application of CNN-based monitoring systems for railway safety.

REFERENCES

- [1] *Rail Freight Transport in the EU: Still Not on the Right Track*, Eur. Court Auditors, Luxembourg 2016.
- [2] B. Tostes, S. T. Henriques, P. E. Brockway, M. K. Heun, T. Domingos, and T. Sousa, "On the right track? Energy use, carbon emissions, and intensities of world rail transportation, 1840–2020," *Appl. Energy*, vol. 367, Jan. 2024, Art. no. 123344.
- [3] A. W. Evans, "The economics of railway safety," *Res. Transp. Econ.*, vol. 43, no. 1, pp. 137–147, Jul. 2013.
- [4] B. Zhang, Q. Yang, F. Chen, and D. Gao, "A real-time foreign object detection method based on deep learning in complex open railway environments," *J. Real-Time Image Process.*, vol. 21, no. 5, 2024, Art. no. 166.
- [5] P. Aela, H.-L. Chi, A. Fares, T. Zayed, and M. Kim, "UAV-based studies in railway infrastructure monitoring," *Autom. Construct.*, vol. 167, Nov. 2024, Art. no. 105714.
- [6] Y. LeCun, Y. Bengio, and G. E. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.

- [7] P. M. Bhatt, R. K. Malhan, P. Rajendran, B. C. Shah, S. Thakar, Y. J. Yoon, and S. K. Gupta, "Image-based surface defect detection using deep learning: A review," *J. Comput. Inf. Sci. Eng.*, vol. 21, no. 4, Aug. 2021, Art. no. 040801.
- [8] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 60, F. Pereira, C. Burges, L. Bottou, and K. Weinberger, Eds., 2017, pp. 84–90.
- [9] K. R. Weiss, T. M. Khoshgoftaar, and D. Wang, "A survey of transfer learning," *J. Big data*, vol. 3, no. 1, pp. 1–40, 2016.
- [10] S. Gao, "A dense attention railway foreign object detection algorithm based on mask R-CNN," *IEEE Access*, vol. 12, pp. 85761–85779, 2024.
- [11] A. Wei and P. Lu, "Research on design of railway supervision and obstacle removal special vehicle based on function analysis method," in *Proc. 5th Int. Conf. Adv. Electron. Mater., Comput. Softw. Eng. (AEMCSE)*, Apr. 2022, pp. 34–39.
- [12] Z. Chen, J. Yang, and F. Zhou, "RailSegVITNet: A lightweight VIT-based real-time track surface segmentation network for improving railroad safety," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 36, no. 1, Jan. 2024, Art. no. 101929.
- [13] R. Kapoor, R. Goel, and A. Sharma, "An intelligent railway surveillance framework based on recognition of object and railway track using deep learning," *Multimedia Tools Appl.*, vol. 81, no. 15, pp. 21083–21109, Jun. 2022.
- [14] M. A. B. Fayyaz and C. Johnson, "Object detection at level crossing using deep learning," *Micromachines*, vol. 11, no. 12, p. 1055, Nov. 2020.
- [15] Y. Yang, Z. Liu, J. Chen, H. Gao, and T. Wang, "Railway foreign object intrusion detection using UAV images and YOLO-UAT," *IEEE Access*, vol. 13, pp. 18498–18514, 2025.
- [16] R. S. Rampriya, T. Al-Shehari, S. Nathan, T. Alfakih, and H. Alsalmán, "An unmanned aerial vehicle captured dataset for railroad segmentation and obstacle detection," *Sci. Data*, vol. 11, no. 1, p. e1315, Dec. 2024.
- [17] F. Wang, Y. Zou, X. Chen, C. Zhang, L. Hou, E. del Rey Castillo, and J. B. P. Lim, "Rapid in-flight image quality check for UAV-enabled bridge inspection," *ISPRS J. Photogramm. Remote Sens.*, vol. 212, pp. 230–250, Jun. 2024.
- [18] *European Railway Safety Statistic*, European Railway Agency, 2023.
- [19] R. Hajovsky, M. Pies, and J. Velicka, "Monitoring the condition of the protective fence above the railway track," *IFAC-PapersOnLine*, vol. 52, no. 27, pp. 145–150, 2019.
- [20] G. Jing, X. Qin, H. Wang, and C. Deng, "Developments, challenges, and perspectives of railway inspection robots," *Autom. Construct.*, vol. 138, Jun. 2022, Art. no. 104242.
- [21] R. Burdzik, B. Nowak, J. Rozmus, P. Słowiński, and J. Pankiewicz, "Safety in the railway industry," *Arch. Transp.*, vol. 44, no. 4, pp. 15–24, 2017.
- [22] Y. LeCun, B. E. Boser, J. S. Denker, D. Henderson, R. Howard, W. E. Hubbard, and L. D. Jackel, "Handwritten digit recognition with a back-propagation network," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 2, 1989, pp. 396–404.
- [23] I. Goodfellow, Y. Bengio, A. Courville, and Y. Bengio, *Deep Learning*, vol. 1. Cambridge, MA, USA: MIT Press, 2016.
- [24] C. Chakraborty, M. Bhattacharya, S. Pal, and S.-S. Lee, "From machine learning to deep learning: Advances of the recent data-driven paradigm shift in medicine and healthcare," *Current Res. Biotechnol.*, vol. 7, Jan. 2024, Art. no. 100164.
- [25] L. Eisentraut, J. Hosch, M. Roytenberg, A. Benecke, P. Penava, and R. Buettner, "Defect detection in industrial soldering processes using machine learning: A critical literature review," *IEEE Access*, vol. 13, pp. 41533–41558, 2025.
- [26] J. Gu, Z. Wang, J. Kuen, L. Ma, A. Shahroudy, B. Shuai, T. Liu, X. Wang, G. Wang, J. Cai, and T. Chen, "Recent advances in convolutional neural networks," *Pattern Recognit.*, vol. 77, pp. 354–377, Jan. 2017.
- [27] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.
- [28] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9.
- [29] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
- [30] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.
- [31] N. Srivastava, G. E. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 1929–1958, 2014.
- [32] F. Zhuang, Z. Qi, K. Duan, D. Xi, Y. Zhu, H. Zhu, H. Xiong, and Q. He, "A comprehensive survey on transfer learning," *Proc. IEEE*, vol. 109, no. 1, pp. 43–76, Jan. 2021.
- [33] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Trans. Knowl. Data Eng.*, vol. 22, no. 10, pp. 1345–1359, Oct. 2009.
- [34] W. Chen, S. Meng, and Y. Jiang, "Foreign object detection in railway images based on an efficient two-stage convolutional neural network," *Comput. Intell. Neurosci.*, vol. 2022, pp. 1–10, Aug. 2022.
- [35] Y. Xu, C. Gao, L. Yuan, S. Tang, and G. Wei, "Real-time obstacle detection over rails using deep convolutional neural network," in *Proc. IEEE Intell. Transp. Syst. Conf. (ITSC)*, Oct. 2019, pp. 1007–1012.
- [36] X. Ding, X. Cai, Z. Zhang, W. Liu, and W. Song, "Railway foreign object intrusion detection based on deep learning," in *Proc. Int. Conf. Comput. Eng. Artif. Intell. (ICCEAI)*, Jul. 2022, pp. 735–739.
- [37] C. Meng, Z. Wang, L. Shi, Y. Gao, Y. Tao, and L. Wei, "SDRC-YOLO: A novel foreign object intrusion detection algorithm in railway scenarios," *Electronics*, vol. 12, no. 5, p. 1256, Mar. 2023.
- [38] W. Zhangyu, Y. Guizhen, W. Xinkai, L. Haoran, and L. Da, "A camera and LiDAR data fusion method for railway object detection," *IEEE Sensors J.*, vol. 21, no. 12, pp. 13442–13452, Jun. 2021.
- [39] Z. Qi, D. Ma, J. Xu, A. Xiang, and H. Qu, "Improved YOLOv5 based on the attention mechanism and FasterNet for foreign object detection on railway and airway tracks," in *Proc. Asian Conf. Commun. Netw. (ASIANComNet)*, Oct. 2024, pp. 1–6.
- [40] T. Ye, Z. Zhao, S. Wang, F. Zhou, and X. Gao, "A stable lightweight and adaptive feature enhanced convolutional neural network for efficient railway transit object detection," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 10, pp. 17952–17963, Oct. 2022.
- [41] D. He, Z. Zou, Y. Chen, B. Liu, X. Yao, and S. Shan, "Obstacle detection of rail transit based on deep learning," *Measurement*, vol. 176, May 2021, Art. no. 109241.
- [42] M. Sevi and İ. Aydın, "Detection of foreign objects around the railway line with YOLOv8," *J. Comput. Sci.*, vol. 23, pp. 19–23, Jan. 2023.
- [43] Y. Wang, L. Wang, Y. H. Hu, and J. Qiu, "RailNet: A segmentation network for railroad detection," *IEEE Access*, vol. 7, pp. 143772–143780, 2019.
- [44] R. S. Rampriya, Sabarinathan, and R. Suganya, "RSNet: Rail semantic segmentation network for extracting aerial railroad images," *J. Intell. Fuzzy Syst.*, vol. 41, no. 2, pp. 4051–4068, Sep. 2021.
- [45] T. Wang, Z. Zhang, F. Yang, and K.-L. Tsui, "Intelligent railway foreign object detection: A semi-supervised convolutional autoencoder based method," 2021, *arXiv:2108.02421*.
- [46] S. Mascarenhas and M. Agarwal, "A comparison between VGG16, VGG19 and ResNet50 architecture frameworks for image classification," in *Proc. Int. Conf. Disruptive Technol. Multi-Disciplinary Res. Appl. (CENTCON)*, vol. 1, Nov. 2021, pp. 96–99.
- [47] L. Eisentraut, C. Mai, J. Hosch, A. Benecke, P. Penava, and R. Buettner, "Deep learning-based detection of tuberculosis using a Gaussian chest X-ray image filter as a software lens," *IEEE Access*, vol. 13, pp. 36065–36081, 2025.
- [48] R. Buettner, C. Mai, and P. Penava, "Improvement of deep learning models using retinal filter: A systematic evaluation of the effect of Gaussian filtering with a focus on industrial inspection data," *IEEE Access*, vol. 13, pp. 43201–43217, 2025.
- [49] G. Deng and L. W. Cahill, "An adaptive Gaussian filter for noise reduction and edge detection," in *Proc. IEEE Conf. Rec. Nucl. Sci. Symp. Med. Imag. Conf.*, Oct. 1993, pp. 1615–1619.
- [50] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, O. Grisel, M. Blondel, P. Prettenhofer, R. J. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, Jan. 2012.
- [51] D. M. W. Powers, "Evaluation: From precision, recall and F-measure to ROC, informedness, markedness and correlation," 2020, *arXiv:2010.16061*.
- [52] M. Grandini, E. Bagli, and G. Visani, "Metrics for multi-class classification: An overview," 2020, *arXiv:2008.05756*.
- [53] M. Sokolova and G. Lapalme, "A systematic analysis of performance measures for classification tasks," *Inf. Process. Manage.*, vol. 45, no. 4, pp. 427–437, Jul. 2009.

- [54] J. Cohen, "A coefficient of agreement for nominal scales," *Educ. Psychol. Meas.*, vol. 20, no. 1, pp. 37–46, Apr. 1960.
- [55] J. Canny, "A computational approach to edge detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vols. PAMI–8, no. 6, pp. 679–698, Nov. 1986.
- [56] F. Guo, J. Liu, Y. Qian, and Q. Xie, "Rail surface defect detection using a transformer-based network," *J. Ind. Inf. Integr.*, vol. 38, Mar. 2024, Art. no. 100584.
- [57] J. Zhao, A. W.-L. Yeung, M. Ali, S. Lai, and V. T.-Y. Ng, "CBAM-SwinT-BL: Small rail surface defect detection method based on Swin transformer with block level CBAM enhancement," *IEEE Access*, vol. 12, pp. 181997–182009, 2024.
- [58] A. Phaphuangwittayakul, N. Harnpornchai, F. Ying, and J. Zhang, "RailTrack-DaViT: A vision transformer-based approach for automated railway track defect detection," *J. Imag.*, vol. 10, no. 8, p. 192, Aug. 2024. [Online]. Available: <https://www.mdpi.com/2313-433X/10/8/192>
- [59] A. Bouguettaya, H. Zarzour, A. Kechida, and A. M. Taberkit, "Deep learning techniques to classify agricultural crops through UAV imagery: A review," *Neural Comput. Appl.*, vol. 34, no. 12, pp. 9511–9536, Jun. 2022.
- [60] Y. Diez, S. Kentsch, M. Fukuda, M. L. L. Caceres, K. Moritake, and M. Cabezas, "Deep learning in forestry using UAV-acquired RGB data: A practical review," *Remote Sens.*, vol. 13, no. 14, p. 2837, Jul. 2021.
- [61] D. Wang, W. Jin, Y. Wu, and J. Ren, "Enhancing adversarial robustness for high-speed train bogie fault diagnosis based on adversarial training and residual perturbation inversion," *IEEE Trans. Ind. Informat.*, vol. 20, no. 5, pp. 7608–7618, May 2024.
- [62] X. Hou, J. An, M. Zhang, B. Du, and J. Liu, "High-speed rail operating environment recognition based on neural network and adversarial training," in *Proc. IEEE 31st Int. Conf. Tools Artif. Intell. (ICTAI)*, Nov. 2019, pp. 840–847.
- [63] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," 2014, *arXiv:1412.6572*.
- [64] Z. Hu, S. Huang, X. Zhu, F. Sun, B. Zhang, and X. Hu, "Adversarial texture for fooling person detectors in the physical world," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13307–13316.
- [65] R. Wiyatno and A. Xu, "Physical adversarial textures that fool visual object tracking," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4821–4830.
- [66] M. Naseer, S. Khan, and F. Porikli, "Local gradients smoothing: Defense against localized adversarial attacks," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2019, pp. 1300–1307.
- [67] D. Ristić-Durrant, M. Franke, and K. Michels, "A review of vision-based on-board obstacle detection and distance estimation in railways," *Sensors*, vol. 21, no. 10, p. 3452, May 2021.
- [68] R. Sañudo, J. Luis Moura, R. Cordera, and A. Ibeas, "Optimal stopping location of a high speed train using GIS and multicriteria decision-making," *Trans. GIS*, vol. 21, no. 1, pp. 151–168, Feb. 2017.
- [69] P. Sikora, L. Malina, M. Kiac, Z. Martinasek, K. Riha, J. Prinosil, L. Jirik, and G. Srivastava, "Artificial intelligence-based surveillance system for railway crossing traffic," *IEEE Sensors J.*, vol. 21, no. 14, pp. 15515–15526, Jul. 2021.
- [70] M. Siergiejczyk, Z. Kasprzyk, M. Rychlicki, and P. Szmigiel, "Analysis and assessment of railway CCTV system operating reliability," *Energies*, vol. 15, no. 5, p. 1701, Feb. 2022.
- [71] C.-T. Lin, S.-W. Huang, Y.-Y. Wu, and S.-H. Lai, "GAN-based day-to-night image style transfer for nighttime vehicle detection," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 2, pp. 951–963, Feb. 2021.
- [72] X. Li, K. Kou, and B. Zhao, "Weather GAN: Multi-domain weather translation using generative adversarial networks," 2021, *arXiv:2103.05422*.



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